1(a)(i)	(halogens can) remove electron(s) (from a reactant)	**1**
1(a)(ii)	$2Al + 3Cl_2 \rightarrow 2AlCl_3$	**1**
1(d)(i)	**M1** $MgCl_2$ (is ionic and) has mobile ions **M2** $SiCl_4$ is covalent **AND** no charge carriers **OR** (no ions and) no delocalised electrons **OR** no mobile / moving ions	**2**
1(d)(ii)	**M1** $MgCl_2$ pH = 7 **M2** $SiCl_4$ pH 0 – 4	**2**

Examiner report missing
9701_w25_qp_24 Question 1

2(a)	**M1** Al_2O_3 is ionic **M2** P_4O_{10} is a (simple) molecule **M3** ionic bonds are stronger than intermolecular forces	**3**

2(c)(iii)	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

Examiner report missing
9701_w25_qp_24 Question 2

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**

1(d)(iii)	**M1** (mixture becomes more) orange **M2** forward reaction is endothermic **M3** (mixture becomes more) orange **M4** equilibrium moves (forwards/right) to compensate for extra acid / H^+	**4**

1(e)	**M1** hydrolysis **M2** $CrF_4 + 2H_2O \rightarrow CrO_2 + 4HF$	**2**

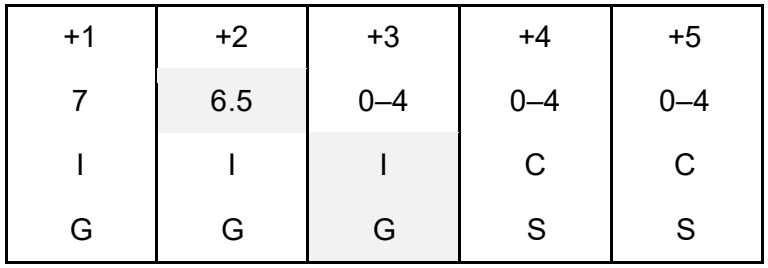
Examiner report missing
9701_w25_qp_23 Question 1

2(a)(i)	**M1** nuclear charge increases (across period) **AND** similar shielding **M2** greater / stronger attraction to nucleus	**2**
2(a)(ii)	**M1** Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– **M2** $Al^{(3+)}$ has (one) fewer shell (than $P^{(3-)}$)	**2**
2(b)(i)		**2**

	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6
M1						
M2	12-14	8-10	—	—	0-4	0-4

2(b)(ii)	they are insoluble	**1**
2(c)(i)	$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$	**1**
2(c)(ii)	$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	**1**

Examiner report missing
9701_w25_qp_23 Question 2

2(a)(i)	delocalised electrons move (through the structure / between layers)	**1**
2(a)(ii)	P / S / C*l* have no charge carriers **OR** (no ions and) no delocalised electrons **OR** no mobile / moving ions	**1**
2(b)	**M1** Si has (lots of) strong covalent bonds broken **M2** P / S / C*l* (only) has weak Van der Waals' (forces) broken	**2**
2(c)		**4**
2(d)(i)	$2Al + 3Cl_2 \rightarrow 2AlCl_3$	**1**
2(d)(ii)	$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$	**1**

Examiner report missing
9701_w25_qp_22 Question 2

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(d)(iii)	**M1** (mixture becomes more) orange **M2** forward reaction is endothermic **M3** (mixture becomes more) orange **M4** equilibrium moves (forwards/right) to compensate for extra acid / H^+	**4**
1(e)	**M1** hydrolysis **M2** $CrF_4 + 2H_2O \rightarrow CrO_2 + 4HF$	**2**

Examiner report missing
9701_w25_qp_21 Question 1

2(a)(i)	**M1** nuclear charge increases (across period) **AND** similar shielding **M2** greater / stronger attraction to nucleus	**2**
2(a)(ii)	**M1** Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– **M2** $Al^{(3+)}$ has (one) fewer shell (than $P^{(3-)}$)	**2**
2(b)(i)	<table><tr><td></td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td><td>(+)6</td></tr><tr><td>**M1**</td><td colspan="6"></td></tr><tr><td>**M2**</td><td>12-14</td><td>8-10</td><td>—</td><td>—</td><td>0-4</td><td>0-4</td></tr></table>	**2**
2(b)(ii)	they are insoluble	**1**
2(c)(i)	$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$	**1**
2(c)(ii)	$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	**1**

Examiner report missing
9701_w25_qp_21 Question 2

1(a)	<table><tr><th>*Mg*</th><th>*Si*</th><th>*P*</th></tr><tr><td>**M1** Giant metallic</td><td>**M2** Giant molecular</td><td>**M3** Simple molecular</td></tr></table>	**3**
1(b)	**M1** Na to A*l* have (delocalised) electrons that can move (through the structure / lattice) **AND** (P / S / C*l*) do not (have delocalised electrons / mobile charge carriers) **M2** Na to A*l* increasing number of delocalised electrons	**2**
1(d)(i)	$P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
1(d)(ii)	$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$	**1**

Examiner report missing
9701_s25_qp_24 Question 1

2(a)(i)	non-linear	**1**
2(b)(i)	*In any order:* **M1** (solution / liquid) (red) to colourless **M2** yellow solid appears	**2**
2(b)(ii)	oxidation number of S in SCl_2 = (+) 2 / (+) II oxidation number of S in H_2SO_3 = (+) 4 / (+) IV	**2**
2(b)(iii)	hydrolysis	**1**

Examiner report missing
9701_s25_qp_24 Question 2

2(a)	Na_2O	**1**
2(b)	$P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$	**1**
2(c)(i)	SO_3	**1**
2(e)(i)	$NaAl(OH)_4$	**1**
2(e)(ii)	$Al_2(SO_4)_3$	**1**

Examiner report missing
9701_s25_qp_23 Question 2

4(a)(i)	Check final graph for range of acceptable answers ● ±(1–0.5) / 100 = ±0.005 ● 0 ● mol dm^{-3} s^{-1} for both Any 2 points = 1 mark, all 3 points = 2 marks	**2**

4(a)(v)	It reacts with water (to produce H_2SO_3)	**1**

Examiner report missing
9701_s25_qp_23 Question 4

1(a)(i)	(both) giant molecular	**1**
1(a)(iii)	no delocalised electrons (to carry charge through the structure) **OR** <u>all</u> (4) outer electrons are used in bonding	**1**
1(b)(i)	$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$	**1**
	$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$	**1**
1(b)(ii)	**M1** phosphorus(V) chloride white / yellow solid **M2** silicon(IV) chloride colourless liquid	**2**
1(b)(iii)	**M1** (phosphorus(V) chloride) produces colourless solution **M2** silicon(IV) chloride produces white suspension / white precipitate (in a colorless liquid / mixture / solution)	**2**
1(c)(i)	species behaves as acid and base **OR** species reacts with (both) acid and base	**1**
1(c)(ii)	Al_2O_3	**1**

Examiner report missing
9701_s25_qp_22 Question 1

<table>
<tr><td>1(a)(i)</td><td>sodium – metallic
AND sodium oxide – ionic</td><td>1</td></tr>
<tr><td>1(a)(iii)</td><td>(sodium and oxide) ions move (through the liquid)</td><td>1</td></tr>
<tr><td>1(b)(i)</td><td>$2Na + 2H_2O \rightarrow 2NaOH + H_2$</td><td>1</td></tr>
<tr><td>1(b)(ii)</td><td>$Na_2O + H_2O \rightarrow 2NaOH$</td><td>1</td></tr>
<tr><td>1(b)(iii)</td><td><table>
<tr><td></td><td>sodium</td><td>sodium oxide</td></tr>
<tr><td>one similarity</td><td colspan="2">(both) solids disappear
OR (both produce) colourless solutions</td></tr>
<tr><td>one difference</td><td>effervescence</td><td>no effervescence</td></tr>
</table></td><td>2</td></tr>
<tr><td>1(c)</td><td>M1 NaCl +1 AND PCl_5 + 5
M2 Na has only 1 valence (shell) electrons AND P has 5</td><td>2</td></tr>
</table>

Examiner report missing
9701_s25_qp_21 Question 1

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(a)(iii)	ratio of moles of P : Cl = $\frac{30.4}{31.0}$: $\frac{69.6}{35.5}$ = 0.98 : 1.96 (= 1 : 2) molecular formula = P_2Cl_4	**2**
1(a)(iv)	(structure =) simple / molecular because low melting point / weak intermolecular forces (bonding =) covalent because hydrolysed	**2**
1(b)(i)	green colour (of gas) disappears	**1**
1(b)(iii)	$P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**

Examiner report missing
9701_m25_qp_22 Question 1

3(a)(i)	<table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table>	**3**
3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	**1**
3(e)(ii)	effervescence / misty / steamy fumes	**1**
3(f)	$Sn(SO_4)_2$	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(a)(iii)	it increases	**1**
	as number of valence electrons (which can be used in bonding / lost / shared / donated) increases	**1**

2(b)(i)

Period 3 oxide	product of rxn with water	pH of solution formed
• MgO	$Mg(OH)_2$	• $8 \leqslant pH \leqslant 12$
P_4O_{10}	• H_3PO_4	• $1 \leqslant pH \leqslant 4$

•✓•✓

2

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

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3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	**1**
3(e)(ii)	effervescence / misty / steamy fumes	**1**
3(f)	$Sn(SO_4)_2$	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

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(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and $SiCl_4$ and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

1(a)(i)	**M1** (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond **M2** (N_2 molecules /molecule(s) of N_2) are non-polar	**2**
1(b)(i)	(+)III	**1**
1(b)(ii)	number of outer / valence electrons	**1**
1(c)(i)	both states shown as solid	**1**
1(c)(ii)	**M1** *correctly balanced equation for aluminium oxide* $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$ **M2** *correctly balanced equation for P_4O_{10} and an excess of NaOH* $P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$	**2**

Question 1

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

(a)(i) This standard recall question could be answered with easy reference to syllabus points.

(ii) With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in N_2 is also straightforward, so the number of σ and π bonds should be readily accessible.

(b)(i) Many candidates answered this question well.

(ii) This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

(c)(i) Many candidates answered this question well.

(ii) The reaction of P_4O_{10} was better known than that of Al_2O_3. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

(d)(i) Many candidates answered this question well.

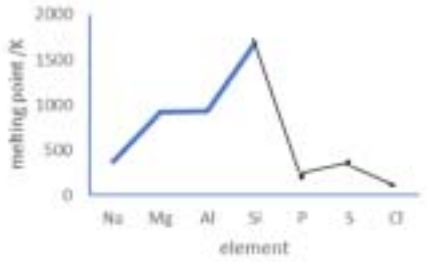
(ii) Many candidates were able to identify the type of reaction that occurs.

1(a)		**2**

Mg^{2+}	24	12	10
Al^{3+}	27	13	10

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(d)(i)	large amount of energy required to break bonds in giant covalent structure	**1**

1(d)(ii)		**2**

M1 all 3 melting points lower than Mg
M2 show melting point Cl < P < S

1(e)(i)		**2**

Na	solid	10–14
S	gas	0–4

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(e)(ii)	acid–base **OR** neutralisation	**1**

1(e)(iii)	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

1(f)(i)	reacts with both acids and bases **OR** shows both acidic and basic behaviour	**1**

1(f)(ii)	$Al(OH)_3 + NaOH \rightarrow NaAl(OH)_4$	**1**

Question 1

(a) The majority of answers were correct.

(b) Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

(c) Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

(d) (i) The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

(ii) Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

(e) (i) This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

(ii) Correct identification of this reaction as acid–base or neutralisation was common.

(iii) The product for the reaction of P_4O_{10} with an excess of water was well known. Equations were not always balanced.

(f) (i) The majority of answers demonstrated a good understanding of the term amphoteric.

(ii) There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

2(a)(i)	hydrolysis	**1**

2(a)(ii)	silicon(IV) chloride / silicon tetrachloride **ALLOW** phosphorus(III) chloride / phosphorus trichloride	**1**

Question 2

(a) (i) This question was well answered.

(ii) The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

(b) (i) The dot-and-cross diagram of SCl_2 was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

(ii) The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in SCl_2, stating that the shape was linear with a bond angle of 180°.

(c) (i) This question was well answered.

(ii) Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

(iii) The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

(d) (i) The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

(ii) Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. SiO_2 and graphene were common incorrect answers.

2(a)	<table><tr><td>formula of chloride</td><td>● $NaCl$</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**
2(c)(i)	● trigonal bipyramidal ● octahedral	**2**
2(c)(ii)	H_2O / water	**1**
	hydrolysis	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
	(vigorous reaction with water suggests) **hydrolysis**	**1**

1(d)(ii)	$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

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A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

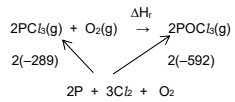
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Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:



ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

1(a)(i)	decreases (from Si to S)	**1**

1(a)(ii)	**M1** Si giant covalent **AND** S is simple molecular **OR** Si giant covalent **AND** S has IMFs	**1**
	M2 Si involves breaking covalent bonds **AND** S involves breaking IMFs	**1**
	M3 more energy needed to break covalent bonds than IMFs **OR** covalent bonds are strong AND IMFs are weak	**1**

1(b)(i)	total e^- in s subshell	6	6	6	**1**
	total e^- in p subshell	8	9	10	**1**

Question 1

(a) (i) The variation in atomic radius from silicon to sulfur was, in general, well known.

(ii) Many candidates appreciated that the differences in melting points were a consequence of the differences in structures. Identification of the types and relative strengths of the forces of attraction broken during melting were referred to less often. Common misconceptions included the presence of intermolecular forces in silicon or did not refer to intermolecular forces, neither in general terms, nor as instantaneous dipole induced dipole attractions in sulfur.

(b) (i) Most candidates correctly calculated the number of electrons in p subshells.

(ii) Many equations showed appropriate species. State symbols, essential in any equation to represent the first ionisation energy process, were included less often.

(iii) Candidates with an excellent knowledge of the trend in ionisation energy across a period identified the correct ionisation energy for phosphorus. They explained both the general trend and the effect of spin pair repulsion on a pair of electrons in a p orbital of sulfur compared to phosphorus. Weaker candidates either identified the value which represents a general trend in ionisation energy across the period and explained this trend or identified the correct value for ionisation energy with details that only explained why sulfur had a lower first ionisation energy than phosphorus.

(iv) This was answered well by most candidates.

(v) Appropriate description of the shape of an SCl_2 molecule as ‘bent’ or ‘v-shape’ was seen often. ‘Linear’ was a common incorrect answer.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
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1(d)(ii)	$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$	**1**

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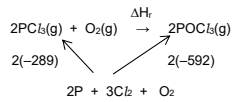
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1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

2(a)(i)	it / aluminium oxide is insoluble (in water) **OR** it / aluminium oxide does not react (with water)	**1**
2(a)(ii)	**M1** correct state of correct aluminium product **M2** balanced equation for reaction of aluminium oxide with NaOH $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4(aq)$ **OR** $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2(aq) + H_2O$	**2**
2(a)(iii)	sodium (only)	**1**
2(c)(i)	Al (+)3 / III **AND** P (+)5 / V	**1**
2(c)(ii)	different number of valence/outer shell electrons	**1**
2(c)(iii)	$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$	**1**
2(c)(iv)	hydrolysis / substitution	**1**
2(c)(v)	HCl(g) dissolves in water to make (hydrochloric) acid	**1**

Question 2

(a) (i) Many candidates knew that aluminium objects covered in a layer of aluminium oxide do not react with cold water because the aluminium oxide is insoluble in water and does not react with the water.

(ii) The equation, with state symbols, was quite well answered, with many candidates recognising that the product was aqueous sodium aluminate.

(iii) Some good answers were seen.

(b) Some candidates gave the formulae of nitrates of elements that are not located in Period 3.

(c) (i) Several candidates correctly stated the oxidation number of Al in $AlCl_3$. The answers for that of P in PCl_5 were varied.

(ii) Many candidates knew that Al and P had different numbers of electrons available to use. Many omitted where these electrons are located.

(iii) It was well recognised that PCl_5 was the required chloride of phosphorus and that HCl is a product of this reaction. Few candidates gave the correct product containing phosphorus. Many answers contained an oxide of phosphorus and/or an incorrect phosphoric acid.

(iv) Some candidates knew the reaction is a hydrolysis reaction with other candidates correctly describing it as a substitution reaction.

(v) Some good answers were seen.

1(a)(i)	bonding in magnesium – metallic **AND** bonding in sodium chloride – ionic	**1**

1(a)(iii)	**M1** S_8 / molecules of sulfur have more electron**s** (than P_4 / molecules of phosphorus) **M2** S has stronger instantaneous dipole–induced dipole forces (than phosphorus / P)	**2**

Question 1

(a) (i) It was generally well known about the bonding present in both species. Occasionally, candidates incorrectly described the bonding in magnesium as ionic.

(ii) The majority of candidates knew that the higher melting point seen for sodium chloride occurred because of stronger ionic bonds. Some candidates identified that there was a difference in bond strength between the two solids without using the information in the table to identify which had the stronger bond.

(iii) Many candidates were able to explain the difference in melting point in terms of the relative strength of intermolecular forces present. Stronger candidates identified the type of intermolecular forces involved and accounted for the difference in terms of the numbers of electrons present in molecules of sulfur and phosphorus. Some candidates incorrectly explained the difference in melting point being due to the difference in strength of covalent bonds.

(b) (i) This definition was generally well known. Some answers were incomplete or did not identify an atom as the appropriate species involved.

(ii) Many candidates knew that the outer electrons become more strongly attracted to the nucleus. Stronger candidates explained this effect in terms of the change in nuclear charge and the effect of shielding across the period.

(iii) It was generally well known that a hydrogen bond is the strongest intermolecular force present between ammonia molecules.

(iv) Stronger candidates were able to correctly show the role of the lone pair of electrons on the nitrogen atom in the formation of a hydrogen bond between ammonia molecules. Of those, some represented the attraction as a dative covalent bond with an arrow from the lone pair. Identification of appropriate partial charges in diagrams was seen more often. A common misconception showed two hydrogen bonds present between a pair of ammonia molecules to represent a dimer.

(v) The majority of candidates described the increased strength of a hydrogen bond in water and many accounted for the increased strength of the hydrogen bond by describing the difference in electronegativity between atoms of nitrogen and oxygen. A much smaller proportion of candidates also correctly referred to the different number of hydrogen bonds per molecule.

2(a)(i)	<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>All correct for one mark	**1**
2(a)(ii)	number of outer/valence electrons	**1**
2(b)(i)	$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$	**1**
2(b)(ii)	1–4	**1**
2(c)(i)	$PCl_5 + 4H_2O \rightarrow 5HCl + H_3PO_4$	**1**
2(c)(ii)	1–4	**1**

Question 2

(a) (i) The maximum oxidation numbers of the Period 3 chlorides were well known.

(ii) The majority of responses were correct. Some candidates produced responses that were vague and incomplete. Occasionally, explanations were incorrectly given in terms of changes to the nuclear charge.

(b) (i) The majority of candidates identified the correct formula of the silicon chloride and gave an appropriate balanced equation for the reaction.

(ii) A significant proportion of candidates incorrectly described the pH of the solution as greater than or equal to 7.

(c) (i) The equation to represent the reaction for phosphorus(V) chloride with water was less well known than the equation for **(b)(ii)**. The production of phosphorus oxide rather than phosphoric acid was a common incorrect answer. Other equations which showed appropriate species were not correctly balanced.

(ii) Many candidates gave appropriate suggestions for the pH of the solution produced in **(c)(i)**.

(d) (i) Equations using C*l* rather than C*l*$_2$ and production of O_2 with HC*l* were common incorrect answers.

(ii) A small proportion of candidates identified all the correct species and produced a balanced equation. Some candidates found the equation for the reaction of chlorine with hot NaOH(aq) to be demanding.

(e) (i) A wide range of incorrect species were seen in the answers; many of which were not ions.

(ii) Many correct equations representing this reaction were seen. Occasionally, equations gave incorrect multiples for the correct species which included the decomposition of some hydrogen peroxide without the use of sodium chlorate(I).

(iii) A significant proportion of candidates used the appropriate data to calculate the number of moles of sodium chlorate(I) used in a single titration. Difficulties arose in attempts to work out the concentration in the original sample, **W**. Occasionally, concentrations were left in mol dm^{-3} rather than converting this into g dm^{-3}, as detailed in the question.

1(a)	<table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table>	**3**
1(f)(i)	$TeF_x + xH_2O \rightarrow Te(OH)_x + xHF$ **OR** $TeF_6 + 6H_2O \rightarrow Te(OH)_6 + 6HF$	**1**
1f(ii)	hydrolysis	**1**

Question 1

(a) This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

(b) Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

(c) Candidates generally answered this well. State symbols were not always included, especially in Te^{+}.

(d) (i) The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

(ii) Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the 6th to 7th ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

(e) Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

(f) (i) This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

(ii) Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

3(b)		$MgCl_2$	PCl_5	**4**
	appearance	● (white) crystals / solid	● (white OR pale yellow) crystals / solid	[1]
	one similarity in observation	(colourless) solution made		[1]
	one difference in observation	no (misty / steamy) fumes	misty / steamy fumes	[1]
	pH	● 6.5–7	● 0–4	[1]

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

1(a)(i)	power of an atom to attract electrons to itself	**1**
1(d)(i)	$SO_2 + H_2O \rightarrow H_2SO_3$	**1**
1(d)(ii)	$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$	**1**
1(d)(iii)	$CO_2 + Mg(OH)_2 \rightarrow MgCO_3 + H_2O$	**1**

Question 1

This question tested ideas of atomic structure, periodicity and some quantitative chemistry. Knowledge of electronegativity was less secure and many Period 3 equations were incorrectly presented as redox reactions.

(a) (i) There were some very good answers here, although many candidates tried to answer in terms of ionisation or electron affinity.

(ii) Few candidates were awarded both marks as this required a balance of points to construct an argument. Spin-pair repulsion arguments were unnecessarily presented on a number of occasions. Many mentioned the distance of the outermost electron from the nucleus without any reference to its significance. Other answers included non-justified references to ionic radius or ionisation energy.

(b) (i) A whole array of different diagrams was seen, with only the ionic form (Li^+ and H^-) acceptable. All electrons, only four in total, were required to be shown.

(ii) Linear, trigonal planar and planar were popular incorrect answers.

(c) (i) The most common error here was the omission of state symbols from the elemental species. Many answers suggested that H_2 was being ionised.

(ii) This was well answered by many candidates.

(iii) Most candidates answered this well, although $3p^6$ was commonly seen.

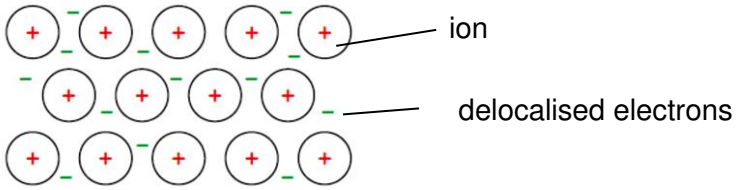
(d) (i) A large number of answers erroneously suggested H_2SO_4 as a product.

(ii) Both Na_2S and Na_2SO_4 appeared regularly, as did H_2. Candidates are advised to consider whether redox has occurred in these reactions.

(iii) Many incorrect versions for the formula of the carbonate were seen such as $Mg(CO_3)_2$ or $MgCO_2$. Many candidates incorrectly gave H_2 as a product from an acid–base reaction.

(e) (i) Many candidates gave the correct overall dipole for S=C=O but then went on to give the remaining three dipoles incorrectly.

(ii) Candidates frequently found it difficult to convert all of the physical quantities to consistent units. Many answers gave the identity as arsenic or some other element and clearly had not used Table 1.2 to identify the compound.

2(a)(i)	 **M1** diagram showing minimum of 4 particles (in total in 2 rows) • circles containing X^{n+} do not have to be labelled •  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled electrons	**1**
	M2 label / legend showing **delocalised** electrons	**1**

2(a)(ii)	Mg has more delocalised e^- (than Na)	**1**

Question 2

This was an inorganic-chemistry question based around Group 2.

(a) (i) A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

(ii) There were many good answers here although some responses continued to use free electrons instead of delocalised.

(b) A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

(c) Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

(d) (i) This question was answered quite well by candidates.

(ii) A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the A_r of $1\times$ or $2 \times Cl$.